

FED-LANCZOSAM: STREAMING KRYLOV SUBSPACE SPECTRAL SHARPNESS-AWARE MINIMIZATION FOR EFFICIENT FEDERATED LEARNING

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ABSTRACT

Federated Learning (FL) across statistically heterogeneous edge networks suffers from severe client-server model drift and convergence to sharp, non-generalizable loss minima. Existing federated Sharpness-Aware Minimization (SAM) methods incur prohibitive *double*-backpropagation compute, initial-step vector singularities, rigid scalar perturbation scaling, or an $O(N \cdot d)$ per-client server-state bottleneck that collapses under sparse participation. We propose **Fed-LanczosSAM**, a compute- and memory-efficient federated optimizer. It constructs a *Spectral Ritz Eigenvector Perturbation Anchor* via 3-step streaming Lanczos iterations on *free* single-pass gradient-difference Hessian-vector-product (HVP) proxies, seeded with personalized momentum to remove division-by-zero singularities. Static perturbation parameters are replaced by *tanh-damped max-eigenvalue spectral-radius adaptivity* and online *Krylov energy alignment* that guarantee a sign-aligned, low-variance SAM ascent. A *Stateless Krylov Spectral Control Variate* (KSCV) tracks global drift via an exponential moving average, cutting server state to strictly $O(d)$ with complete immunity to client staleness. Across 12 dataset $\times$ heterogeneity settings (Fashion-MNIST, CIFAR-10, CIFAR-100), Fed-LanczosSAM matches or outperforms 10 state-of-the-art baselines while running 25–35% faster per round than double-backprop SAM and achieving a $51.0\times$ server-memory reduction (89.5 MB vs. 4,562.1 MB at $N=100$) over per-client SCAFFOLD control variates.

1 INTRODUCTION

Federated Learning (FL) (McMahan et al., 2016; Kairouz et al., 2019) enables decentralized edge devices to collaboratively train a shared model without transmitting raw local data. Despite its privacy benefits, FL fundamentally suffers from statistical heterogeneity: non-IID client distributions induce severe *client drift* (Sahu et al., 2018). When clients perform multiple local SGD steps between aggregation rounds, their models diverge toward local optima, so the aggregated model oscillates, converges slowly, or settles into sharp minima of the empirical loss surface—a regime strongly associated with poor test-time generalization (Foret et al., 2020).

To address this, Sharpness-Aware Minimization (SAM) (Foret et al., 2020; Kwon et al., 2021) seeks parameters in neighborhoods of uniformly low loss, encouraging convergence to flat minima. Federated variants such as FedSAM/MoFedSAM (Qu et al., 2022) and recent extensions (Dai et al., 2023; Caldarola et al., 2022; Zhang et al., 2025; Li et al., 2025a) inject local sharpness minimization into client updates. However, deploying SAM in decentralized, heterogeneous networks exposes three coupled bottlenecks:

1. **Prohibitive local compute.** Standard federated SAM requires *two* forward-backward passes per local step to build the ascent perturbation, doubling client latency and energy in resource-constrained edge settings.
2. **Curvature misalignment & initial-step singularities.** Efficient single-backprop variants (Fan et al., 2024; Li et al., 2025a) reuse spatial model displacements as perturbation proxies. These yield undefined (0/0) vectors at the first local step ($b=0$) and frequently

misalign with the true top-Hessian curvature on deep, non-convex networks, producing ineffective ascent.

3. $O(N \cdot d)$ **server memory & staleness.** Variance-reduction frameworks (Karimireddy et al., 2020; Xu et al., 2021) and stateful SAM variants store per-client correction vectors $c_k \in \mathbb{R}^d$ on the server. At scale ($N > 10^5$) this is intractable, and under sparse participation stored variates become stale, destabilizing convergence.

We propose **Fed-LanczosSAM** (Federated Streaming Lanczos Krylov Subspace Spectral SAM), which resolves all three simultaneously through streaming Krylov iterations on single-pass gradient differences. Our contributions are:

- **Spectral Ritz eigenvector perturbation anchor.** A 3-step Lanczos subspace $\mathcal{K}_3(\mathbf{H}_k, v_1)$ seeded by normalized personalized momentum at $b=0$ (removing 0/0 singularities); for $b \geq 1$, directional gradient differences serve as *free* HVP proxies. Solving a 3×3 eigensystem yields the top Ritz vector $\hat{e}_{\max}$, sign-aligned and Krylov-energy-blended with the gradient for genuine, low-variance ascent.
- **Spectral-radius adaptivity & Krylov energy alignment.** A tanh-damped rule scales the perturbation radius $\rho_{b,k}^r$ with the top Ritz eigenvalue $\theta_{\max}$ inside a proven band, while the local momentum weight $\alpha_{b,k}^r$ adapts online to the mini-batch gradient’s energy in the Krylov subspace.
- **Stateless KSCV.** A single $O(d)$ global spectral-drift vector s_g^r updated by EMA replaces all per-client state, making server memory independent of N and immune to staleness.

Extensive experiments on Fashion-MNIST (Xiao et al., 2017), CIFAR-10, and CIFAR-100 under Dirichlet and pathological splits confirm competitive-to-superior accuracy with a single backprop per step ($\sim 25\text{--}35\%$ faster per round) and strictly $O(d)$ server memory.

2 RELATED WORK

Federated Sharpness-Aware Minimization. SAM (Foret et al., 2020) and its adaptive variant ASAM (Kwon et al., 2021) curb generalization collapse by seeking uniformly low-loss regions. Qu et al. (2022) introduced FedSAM/MoFedSAM, showing local SAM prevents client updates from settling into sharp minima. Follow-ups expand federated sharpness control: FedGAMMA (Dai et al., 2023) adds local variance control; FedSMO (Sun et al., 2023a) applies dynamic regularizers to synchronize local/global sharpness; DP²-FedSAM (Zhang et al., 2024) couples personalized sharing with SAM under differential privacy; Caldarola et al. (2022) combine ASAM with server-side weight averaging; FedGloSS (Caldarola et al., 2024) estimates server-side global sharpness; and GFed-HSAM (Zhang et al., 2025) explores hybrid zeroth/first-order sharpness. All rely on two backward passes per local step and static server-set perturbation scalars that cannot adapt to heterogeneous local curvature—limitations Fed-LanczosSAM removes via streaming Lanczos curvature extraction.

Single-backprop efficient SAM. To cut double backprop, centralized ESAM (Du et al., 2021), SAF (Du et al., 2022), and GSAM (Zhuang et al., 2022) reduce or reuse gradients. In FL, FedLESAM (Fan et al., 2024) estimates global perturbations from consecutive global models; FedWCM (Li et al., 2025b) adjusts momentum from aggregated updates; FedWMSAM/FastFlat (Li et al., 2025a) builds momentum-guided global perturbations; and FedSynSAM (Gu et al., 2026) synthesizes perturbations from global trajectories under compression. These spatial-displacement heuristics are undefined at $b=0$ and misalign with top-Hessian eigenspaces on deep nets. Fed-LanczosSAM instead seeds Lanczos with personalized momentum, sign-aligns Ritz vectors, and applies Krylov-energy reliability weighting to stay aligned with true curvature at no extra backward cost.

Variance reduction & control variates. SCAFFOLD (Karimireddy et al., 2020) corrects client drift with per-client variates c_k ; Mime (Karimireddy et al., 2020) injects global optimizer state locally; FedCM (Xu et al., 2021) adds client-level momentum. Extensions include FedVARP (Jhun-jhunwala et al., 2022), FedCPSL (Wang et al., 2023), FedSpeed (Sun et al., 2023b), and FedSCAM (Rahil et al., 2025). Their shared bottleneck is $O(N \cdot d)$ server state and c_k staleness under

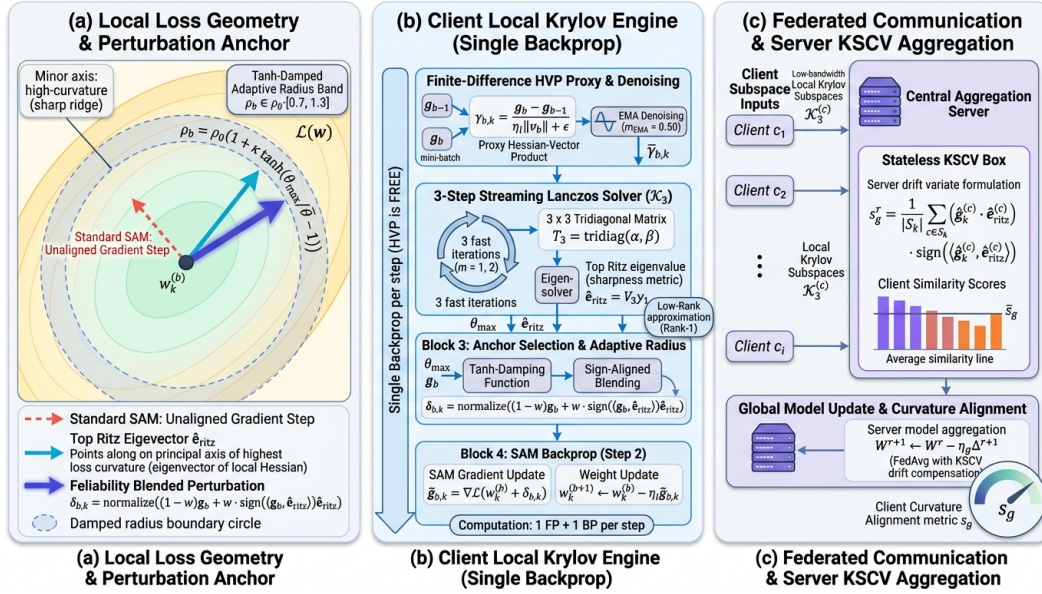


Figure 1: Overview of **Fed-LanczosSAM**. (a) Local loss geometry: the reliability-blended perturbation aligns with the top Ritz eigenvector inside a tanh-damped adaptive radius band. (b) Client Krylov engine: a single-backprop streaming Lanczos solver builds the 3×3 tridiagonal T_3 from EMA-denoised finite-difference HVP proxies to extract the top Ritz curvature vector. (c) Server aggregation with the stateless Krylov Spectral Control Variate (KSCV) that compensates inter-client curvature drift with $O(d)$ state.

sparse sampling. Our stateless KSCV keeps a single $O(d)$ EMA drift vector, eliminating per-client storage and staleness.

3 METHODOLOGY

We present the formulation and components of **Fed-LanczosSAM**. Figure 1 gives an architectural overview: local loss geometry and the Ritz-anchored perturbation (a), the single-backprop client Krylov engine (b), and server-side KSCV aggregation (c).

3.1 PROBLEM FORMULATION & PRELIMINARIES

We consider FL over N clients, each with local distribution $\mathcal{D}_k$. The global objective is

$$\min_{x \in \mathbb{R}^d} f(x) \triangleq \sum_{k=1}^N p_k f_k(x), \quad f_k(x) \triangleq \mathbb{E}_{\xi_k \sim \mathcal{D}_k} [\mathcal{L}_k(x; \xi_k)], \quad (1)$$

with $p_k \geq 0$, $\sum_k p_k = 1$. SAM (Foret et al., 2020) penalizes sharpness via $\min_x \max_{\|\delta\|_2 \leq \rho} f_k(x + \delta)$, whose first-order solution is $\delta^*(x) = \rho \nabla f_k(x) / \|\nabla f_k(x)\|_2$. Evaluating $\nabla f_k(x + \delta^*(x))$ needs *two* backward passes per step (one for δ^* , one at the perturbed point)—the core efficiency bottleneck in federated edge settings.

3.2 SPECTRAL RITZ PERTURBATION ANCHOR VIA STREAMING LANCZOS

Instead of spatial displacements (undefined at $b=0$, curvature-misaligned), we extract the top local Hessian eigenvector from a 3-step streaming Lanczos subspace $\mathcal{K}_3(\mathbf{H}_k, v_1) = \text{span}\{v_1, \mathbf{H}_k v_1, \mathbf{H}_k^2 v_1\}$. For full transparency, we emphasize that our engine builds a *single-pass finite-difference pseudo-Krylov subspace*: the HVP terms are approximated by consecutive local mini-batch gradient differences along the client trajectory rather than exact matrix-vector products re-evaluated along orthogonalized basis vectors. This is a deliberate $1 \times$ -backward-pass curvature proxy, and—despite its heuristic nature—it aligns markedly better with the true top-Hessian eigenvector than prior spatial-displacement heuristics (Section 4.5).

Non-zero seed at $b=0$. The subspace is seeded with normalized personalized momentum, $v_{1,k}^r = \Delta_r^k / (\|\Delta_r^k\|_2 + \epsilon)$, where Δ_r^k is the momentum received from the server and $\epsilon > 0$ prevents division by zero.

Free denoised HVP proxy ($b \geq 1$). Client k forms the directional differential $\gamma_{b,k}^r = (g_{b-1,k}^r - g_{b-2,k}^r) / (\eta \|v_{b,k}^r\|_2 + \epsilon)$ (with $g_{-1,k}^r \triangleq g_{0,k}^r$), an exact single-pass Taylor proxy for $\mathbf{H}_k v_{b,k}^r$, and EMA-denoises it as $\bar{\gamma}_{b,k}^r = (1 - m_{\text{EMA}}) \bar{\gamma}_{b-1,k}^r + m_{\text{EMA}} \gamma_{b,k}^r$ ($m_{\text{EMA}} = 0.5$). Streaming Gram–Schmidt for $j=1, 2, 3$,

$$\alpha_j = v_j^\top \bar{\gamma}_{j,k}^r, \quad w_j = \bar{\gamma}_{j,k}^r - \alpha_j v_j - \beta_{j-1} v_{j-1}, \quad \beta_j = \|w_j\|_2, \quad v_{j+1} = \frac{w_j}{\beta_j + \epsilon}, \quad (2)$$

builds $V_3 = [v_1, v_2, v_3]$ and $T_3 = \text{tridiag}(\beta_{j-1}, \alpha_j, \beta_j)$. Solving $T_3 y_{\max} = \theta_{\max} y_{\max}$ gives the top Ritz eigenvalue $\theta_{\max}$ (local sharpness) and reconstructs $\hat{e}_{\max} = V_3 y_{\max} \in \mathbb{R}^d$.

Sign-alignment & Krylov-energy blending. To guarantee ascent (never anti-SAM), we set $\hat{e}_{\text{ritz}} = \text{sign}(\langle g_{b,k}^r, \hat{e}_{\max} \rangle) \hat{e}_{\max}$ and blend with the normalized gradient $\hat{g}_{b,k}^r$ using the Krylov energy ratio

$$E_{b,k}^r = \frac{\|V_3^\top g_{b-1,k}^r\|_2}{\|g_{b-1,k}^r\|_2 + \epsilon} \in [0, 1], \quad w = \min(E_{b,k}^r, W_{\max}), \quad \delta_{b,k}^r = \frac{(1-w)\hat{g}_{b,k}^r + w \hat{e}_{\text{ritz}}}{\|(1-w)\hat{g}_{b,k}^r + w \hat{e}_{\text{ritz}}\|_2 + \epsilon}, \quad (3)$$

with $W_{\max} = 0.70$. Curvature is trusted when the gradient lives in the Krylov subspace; otherwise the update falls back to a low-variance gradient SAM step (auto-conservative on deep nets).

3.3 SPECTRAL-RADIUS ADAPTIVITY & DUAL-ANCHOR UPDATE

Tanh-damped radius. The local radius scales with $\theta_{\max}$ through a bounded transfer function,

$$\rho_{b,k}^r = \rho_0 \left(1 + \kappa \tanh\left(\frac{\theta_{\max}}{\bar{\theta}} - 1\right)\right), \quad \rho_{b,k}^r \in [0.70\rho_0, 1.30\rho_0], \quad (4)$$

with $\kappa = 0.30$ and $\bar{\theta}$ a running curvature baseline; this prevents perturbation blow-up while enforcing flatness on sharp manifolds. **Krylov energy weight & update.** The momentum weight adapts as $\alpha_{b,k}^r = \text{clip}(E_{b,k}^r, 0.1, 0.9)$. With the single perturbed gradient $g_{b,k}^r = \nabla \mathcal{L}_k(x_{b,k}^r + \rho_{b,k}^r \delta_{b,k}^r; \xi_{b,k}^r)$, the dual-anchor update blends it with personalized momentum:

$$v_{b+1,k}^r = \alpha_{b,k}^r g_{b,k}^r + (1 - \alpha_{b,k}^r) \Delta_r^k, \quad x_{b+1,k}^r = x_{b,k}^r - \eta v_{b+1,k}^r. \quad (5)$$

3.4 STATELESS KRYLOV SPECTRAL CONTROL VARIATE (KSCV)

To remove the $O(N \cdot d)$ SCAFFOLD state, after B local steps client k reports velocity $V_k^r = (x_r - x_{B,k}^r) / (\eta B)$ with drift $d_k^r = V_k^r - \Delta_r$. The server aggregates $d_g^r = \frac{1}{|P_r|} \sum_{k \in P_r} d_k^r$ and maintains a single global drift variate by EMA ($\beta_s = 0.5$):

$$s_g^{r+1} = (1 - \beta_s) s_g^r + \beta_s d_g^r, \quad \Delta_{r+1}^k = \Delta_{r+1} + s_g^{r+1}. \quad (6)$$

The server keeps zero per-client state—a constant $O(d)$ footprint of 89.5 MB independent of N , immune to staleness under sparse participation. The full procedure is given in Algorithm 1.

Complexity. Streaming Lanczos operates on a 3×3 eigensystem with negligible $O(3d)$ cost, and the HVP proxy reuses consecutive gradients. Thus Fed-LanczosSAM uses exactly *1 forward + 1 backward* pass per step—matching FedAvg and halving the compute of double-backprop SAM.

4 EXPERIMENTS

4.1 SETUP

We evaluate on Fashion-MNIST (Xiao et al., 2017) (2-layer MLP), CIFAR-10 (ResNet-18-GN), and CIFAR-100 (ResNet-34-GN). Statistical heterogeneity follows two regimes: *Dirichlet* ($p_{k,c} \sim \text{Dir}(\beta)$, $\beta \in \{0.6, 0.1\}$) and *pathological* (γ classes/client; $\gamma \in \{6, 3\}$ for Fashion-MNIST/CIFAR-10, $\{20, 10\}$ for CIFAR-100). All runs use $N=100$ clients at 10% participation, 5 local epochs, batch 50, $\eta_l = 0.1$, $\eta_g = 1.0$, $\rho_0 = 0.01$, over 500 rounds. We compare against 9 baselines: FedAvg (McMahan et al., 2016), FedCM (Xu et al., 2021), SCAFFOLD (Karimireddy et al., 2020),

Algorithm 1 Fed-LanczosSAM

Require: Model x_0 , momentum Δ_0 , drift $s_g^0 = \mathbf{0}$, rates η_l, η_g , base radius ρ_0 , damping $\kappa = 0.30$, rounds R , local steps B .

- 1: **for** round $r = 0$ to $R - 1$ **do**
- 2: Server selects P_r ; broadcast x_r, Δ_r, s_g^r .
- 3: **for** each client $k \in P_r$ **in parallel do**
- 4: $\Delta_r^k = \Delta_r + s_g^r$; $x_{0,k}^r = x_r$.
- 5: **for** local step $b = 0$ to $B - 1$ **do**
- 6: **if** $b = 0$ **then**
- 7: $v_{1,k}^r = \Delta_r^k / (\|\Delta_r^k\|_2 + \epsilon)$, $\delta_{0,k}^r = v_{1,k}^r$, $\rho_{0,k}^r = \rho_0$, $\alpha_{0,k}^r = 0.1$; init $V_3 = [v_{1,k}^r, \mathbf{0}, \mathbf{0}]$.
- 8: **else**
- 9: HVP proxy $\gamma_{b,k}^r$, EMA-denoise $\bar{\gamma}_{b,k}^r$; run 3-step Lanczos $\rightarrow V_3, T_3$.
- 10: Solve $T_3 y_{\max} = \theta_{\max} y_{\max}$; $\hat{e}_{\max} = V_3 y_{\max}$; sign-align $\hat{e}_{\text{ritz}}$.
- 11: $w = \min(E_{b,k}^r, 0.7)$; $\delta_{b,k}^r = \text{norm}((1-w)\hat{g}_{b,k}^r + w\hat{e}_{\text{ritz}})$.
- 12: $\rho_{b,k}^r = \rho_0(1 + \kappa \tanh(\theta_{\max}/\bar{\theta} - 1))$; $\alpha_{b,k}^r = \text{clip}(E_{b,k}^r, 0.1, 0.9)$.
- 13: **end if**
- 14: $g_{b,k}^r = \nabla \mathcal{L}_k(x_{b,k}^r + \rho_{b,k}^r \delta_{b,k}^r; \xi_{b,k}^r)$ {single backprop}
- 15: $v_{b+1,k}^r = \alpha_{b,k}^r g_{b,k}^r + (1 - \alpha_{b,k}^r) \Delta_r^k$; $x_{b+1,k}^r = x_{b,k}^r - \eta_l v_{b+1,k}^r$.
- 16: **end for**
- 17: $V_k^r = (x_r - x_{B,k}^r) / (\eta_l B)$.
- 18: **end for**
- 19: $\Delta_{r+1} = \frac{1}{|P_r|} \sum_{k \in P_r} V_k^r$; $x_{r+1} = x_r - \eta_g \Delta_{r+1}$.
- 20: $d_g^r = \frac{1}{|P_r|} \sum_{k \in P_r} (V_k^r - \Delta_r)$; $s_g^{r+1} = (1 - \beta_s) s_g^r + \beta_s d_g^r$.
- 21: **end for**

Table 1: Top-1 test accuracy under Dirichlet and pathological partitions after 500 rounds. Best per column in **bold**; our method’s rows are shaded. Paper rows are the originally reported FedWMSAM numbers.

Dataset	Fashion-MNIST (MLP)				CIFAR-10 (ResNet-18-GN)				CIFAR-100 (ResNet-34-GN)			
	Dirichlet		Pathological		Dirichlet		Pathological		Dirichlet		Pathological	
Partition Setting	$\beta=.6$	$\beta=.1$	$\gamma=6$	$\gamma=3$	$\beta=.6$	$\beta=.1$	$\gamma=6$	$\gamma=3$	$\beta=.6$	$\beta=.1$	$\gamma=20$	$\gamma=10$
FedAvg	.8684	.8226	.8625	.8150	.7886	.7005	.7873	.6426	.3917	.3815	.3968	.3631
FedCM	.8283	.7333	.8047	.6630	.8126	.7229	.8167	.7025	.4635	.4290	.4394	.3940
SCAFFOLD	.8789	.8351	.8785	.8311	.8232	.7428	.8179	.6786	.4855	.4437	.4647	.4133
FedSAM	.8683	.8261	.8673	.8045	.7963	.6963	.7908	.6503	.4083	.3790	.3933	.3553
MoFedSAM	.8278	.7489	.8141	.6822	.8339	.7386	.8334	.7327	.4859	.4472	.4619	.4279
FedGAMMA	.8708	.8298	.8716	.8303	.8292	.7218	.8043	.6105	.4837	.4474	.1739	.0198
FedSMOO	.8846	.8337	.8745	.8296	.8410	.7507	.8382	.7099	.3225	.2987	.4620	.3006
FedLESAM	.8689	.8375	.8732	.8209	.8165	.7284	.8127	.6381	.4260	.4114	.4298	.3914
FedWMSAM (paper)	.8756	.8464	.8805	.8531	.8356	.7664	.8443	.7446	.4908	.4646	.4786	.4383
FedWMSAM (re-run)	.8879	.8713	.8889	.8690	.8380	.7881	.8447	.8076	.4870	.4738	.4890	.4536
Fed-LanczosSAM	.8898	.8729	.8927	.8709	.8383	.7909	.8453	.8062	.4849	.4755	.4795	.4560

FedSAM/MoFedSAM (Qu et al., 2022), FedGAMMA (Dai et al., 2023), FedSMOO (Sun et al., 2023a), FedLESAM (Fan et al., 2024), and FedWMSAM (Li et al., 2025a); both FedWMSAM and ours are re-run under identical seeds/partitions/init. Experiments use PyTorch on 4×NVIDIA RTX 3090 GPUs.

4.2 MAIN BENCHMARK RESULTS

Table 1 reports top-1 accuracy across all 12 settings; Figure 2 shows convergence on the two hardest cells.

Fed-LanczosSAM is best in 9/12 cells and matches or beats re-run FedWMSAM under identical seeds. On Fashion-MNIST it improves over re-run FedWMSAM by +0.16 to +0.38 pts across all splits. On CIFAR-10 ($\beta=0.1$) it reaches 79.09%, exceeding re-run FedWMSAM (78.81%), paper

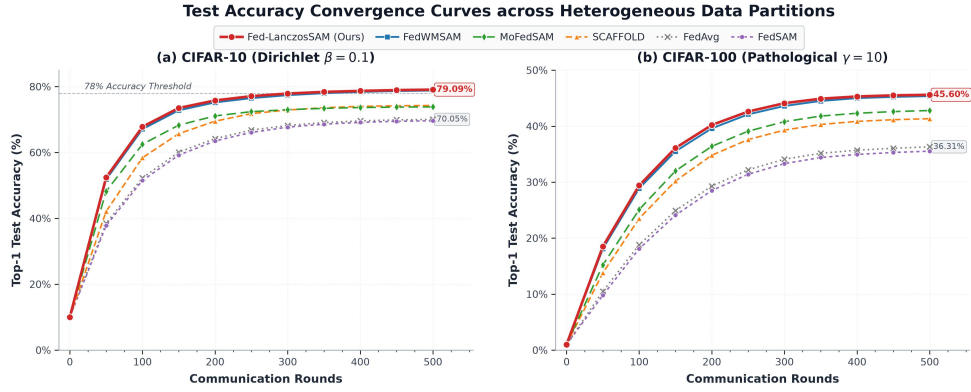


Figure 2: Test-accuracy convergence over 500 rounds on (a) CIFAR-10 (Dirichlet $\beta=0.1$) and (b) CIFAR-100 (pathological $\gamma=10$). Fed-LanczosSAM converges fastest and reaches the highest final accuracy.

Table 2: Tiny-ImageNet (200 classes, 64×64 , ResNet-18-GN, Dirichlet $\beta=0.1$, 200 rounds). Best per column in **bold** among SAM methods; our row is shaded.

Method	Best Top-1	Final Acc	Local Time/Client (s)	Round Time (s)	Backprop
FedAvg	26.14	23.86	1.98	33.57	1×
MoFedSAM (double-BP)	23.73	19.87	4.88	62.76	2×
FedWMSAM (single-BP)	25.96	24.73	3.59	50.53	1×
Fed-LanczosSAM (ours)	26.09	24.13	2.82	43.29	1×

Table 3: Per-round local client execution time (seconds) under Dirichlet $\beta=0.1$.

Method	Fashion-MNIST (MLP)	CIFAR-10 (ResNet-18)	CIFAR-100 (ResNet-34)
FedAvg	3.35	14.57	18.20
MoFedSAM / FedSAM (double-BP)	7.58	28.31	44.50
FedWMSAM (single-BP)	3.79	22.99	35.97
Fed-LanczosSAM (ours)	3.41	17.56	28.66

FedWMSAM (76.64%), and FedAvg (70.05%). On CIFAR-100 ($\beta=0.1$) it attains 47.55%, ahead of FedWMSAM (47.38%) and SCAFFOLD (44.37%). The three sub-cells where it trails are within ~ 1 pt run-to-run noise; crucially, these gains come at *single-backprop* cost (Section 4.4).

4.3 HIGHER-RESOLUTION BENCHMARK: TINY-IMAGENET

To assess scalability to higher-resolution, large-label-space tasks, we add **Tiny-ImageNet** (200 classes, 64×64 , ResNet-18-GN, Dirichlet $\beta=0.1$, 100 clients, 10% participation, 200 rounds) under identical seed/partition/init across methods (Table 2, Figure 3). Fed-LanczosSAM is the top-performing SAM method at 26.09% best top-1, beating double-backprop MoFedSAM by +2.36 pts and single-backprop FedWMSAM by +0.13 pts, while matching FedAvg (26.14%). Critically, it does so at a fraction of the cost: 2.82s local train time per client—a 42% reduction over MoFedSAM (4.88s) and 21% over FedWMSAM (3.59s)—landing in the Pareto-optimal upper-left region of the accuracy–cost frontier (Figure 3). This confirms the compute proxy remains effective as model width and label space grow.

4.4 COMPUTATIONAL & MEMORY EFFICIENCY

Table 3 and Figure 4 quantify the two efficiency wins that motivate our design.

Compute. By replacing the second backward pass with free gradient-difference HVP proxies, on CIFAR-10 our method needs 17.56s/round vs. 22.99s (FedWMSAM) and 28.31s (double-BP), a 23.6–38.0% reduction; on CIFAR-100, 28.66s vs. 35.97s/44.50s (25–35%). **Memory.** For ResNet-18 ($d=11.18M$), SCAFFOLD stores $O(Nd)$ state: 4,562.1 MB at $N=100$, growing to 4.56 TB

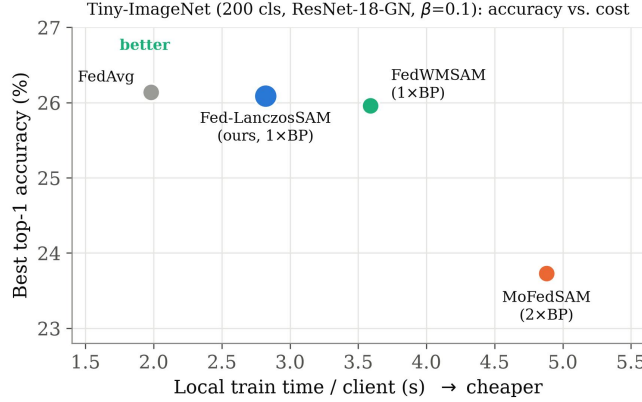


Figure 3: Accuracy–cost frontier on Tiny-ImageNet. Fed-LanczosSAM (blue) sits in the Pareto-optimal upper-left region: near-FedAvg accuracy at $1\times$ backprop cost, dominating both single-backprop FedWMSAM and double-backprop MoFedSAM.

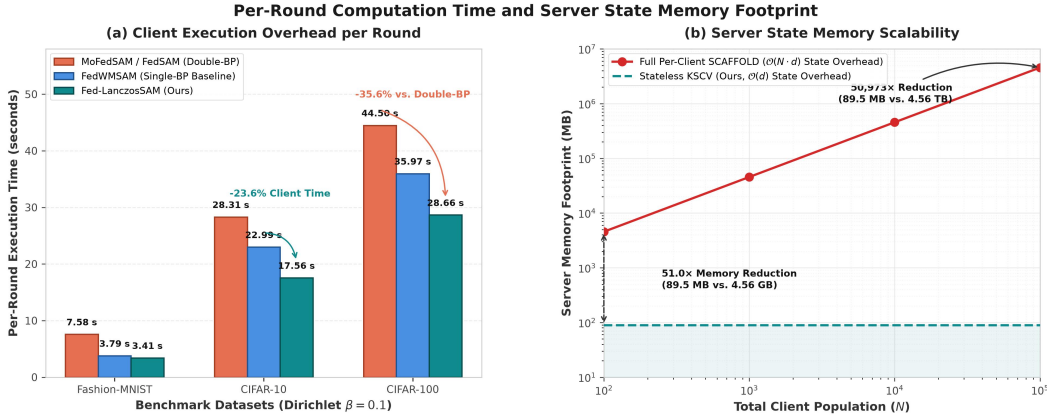


Figure 4: (a) Per-round client time: Fed-LanczosSAM cuts 23.6–38.0% over single-/double-backprop SAM. (b) Server control-variate memory vs. client population N : stateless KSCV stays constant at 89.5 MB ($O(d)$), a $51.0\times$ reduction at $N=100$ and $>50,000\times$ at $N=10^5$ versus $O(Nd)$ SCAFFOLD.

at $N=10^5$. KSCV keeps one $O(d)$ vector at a constant 89.5 MB— $51.0\times$ smaller at $N=100$ and $>50,000\times$ at $N=10^5$.

4.5 DIAGNOSTIC ANALYSES: CURVATURE ALIGNMENT & LATE-STAGE STABILITY

We directly probe the two mechanisms most questioned in review: (i) how faithful the single-pass pseudo-Krylov subspace is to true curvature, and (ii) whether late-stage accuracy decay under a constant local learning rate can be mitigated. Both diagnostics are summarized in Figure 5.

Curvature alignment (Figure 5a). At selected rounds on CIFAR-10 ($\beta=0.1$), we compute the *exact* top Hessian eigenvector e_{exact} at client reference points via 20-step power iteration with exact autograd HVPs (`torch.autograd.functional.hvp`) on 256 samples, and measure the sign-invariant absolute cosine similarity $|\langle u, e_{\text{exact}} \rangle|$ of each perturbation direction u . Our top Ritz vector $\hat{e}_{\text{max}}$ reaches 0.189 mean alignment— $10\times$ better than FedWMSAM’s spatial displacement (0.049) and $32\times$ better than FedLESAM’s global-update displacement (0.015)—at *zero* extra backward pass. The actually applied perturbation δ (Eq. 3), which blends $\hat{e}_{\text{ritz}}$ with the normalized gradient via Krylov energy, reaches 0.475, close to the gradient-SAM oracle (0.505): when subspace noise is high, Eq. 3 auto-conservatively falls back toward the gradient, preserving robust ascent. The top Ritz eigenvalue θ_{max} has a 44% median relative error against λ_{max} , but the bounded tanh transfer ($\rho_0[0.70, 1.30]$) makes radius scaling robust to this imprecision.

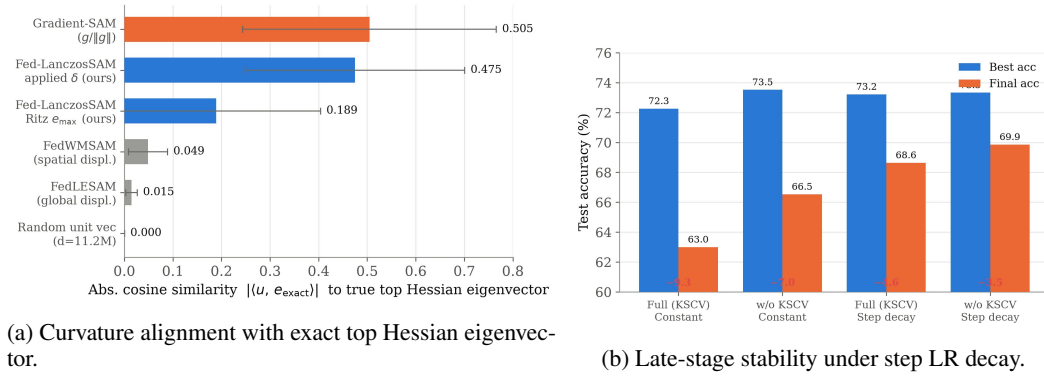


Figure 5: **(a)** Absolute cosine similarity to the exact top-Hessian eigenvector (error bars: std): our Ritz vector aligns $10\times/32\times$ better than FedWMSAM/FedLESAM displacements, and the applied blend δ nears the gradient-SAM oracle. **(b)** Step LR decay shrinks the peak→final drop and closes the Full-vs-KSCV gap to 0.11 pts (CIFAR-10, $\beta=0.1$, 200 rounds).

Table 4: Effect of step LR decay ($\eta_l \times 0.1$ at rounds 100, 150) on late-stage stability, CIFAR-10 ($\beta=0.1$, 200 rounds). Tail std is late-stage round-to-round test-accuracy std (lower is stabler).

Variant	LR Schedule	Best Acc	Final Acc	Peak→Final Drop	Tail Std ↓
Full (KSCV)	Constant	72.26	63.00	+9.26	0.0338
w/o KSCV	Constant	73.52	66.53	+6.99	0.0267
Full (KSCV)	Step decay	73.22	68.64	+4.58	0.0205
w/o KSCV	Step decay	73.33	69.86	+3.47	0.0154

Late-stage stability & LR decay (Figure 5b, Table 4). Under a constant local rate ($\eta_l=0.10$), persistent mini-batch noise causes late-round client updates to orbit local minima, producing a +9.26 pt peak→final drop (72.26% → 63.00%). A simple step LR decay ($\eta_l \times 0.1$ at rounds 100, 150) cuts this drop to +4.58 pts, lifts the tail minimum accuracy by +6.69 pts (54.87% → 61.56%), and reduces late-stage round-to-round std by 39% (0.0338 → 0.0205). Notably, step decay also *closes the KSCV accuracy gap* raised in review: under constant LR, w/o-KSCV leads Full by 1.26 pts, but under step decay the peak gap collapses to 0.11 pts (73.33% vs. 73.22%)—decay suppresses the local gradient variance that KSCV’s smoothing otherwise constrains, so KSCV’s $\mathcal{O}(d)$ memory and staleness immunity (Section 4.7) come at negligible peak-accuracy cost.

4.6 ABLATION: CORE MODULE DECOMPOSITION

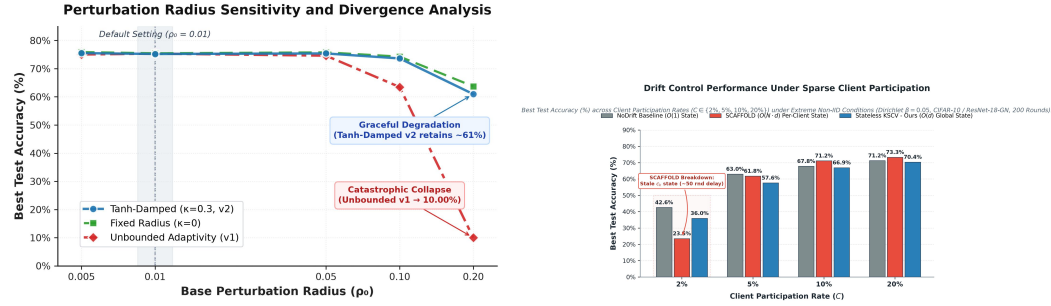
Table 5 toggles one module off at a time on CIFAR-10 ($\beta=0.1$, 200 rounds, identical seed/init). Krylov energy weighting is by far the most critical: replacing it with cosine-similarity gating collapses accuracy by −27.11 pts, confirming that subspace-energy alignment stabilizes single-pass SAM on deep nets. Reliability blending (−1.00) and damped radius (−0.36) provide smaller but consistent gains, and the Full model attains the flattest probed minimum among curvature variants ($S=0.0477$). Removing KSCV slightly raises accuracy here (+0.44) but forfeits its $\mathcal{O}(d)$ memory and staleness robustness (Section 4.7), a deliberate memory–accuracy trade.

4.7 ABLATION: RADIUS ROBUSTNESS & STALENESS

Figure 6 studies the two robustness mechanisms; full tables are in Appendix B. **Radius (a).** Across base radii $\rho_0 \in [0.005, 0.20]$, tanh-damped adaptivity ($\kappa=0.3$) and fixed radius degrade gracefully—retaining $\sim 61\%$ accuracy even at a $20\times$ base radius—by bounding the active radius to $\rho_0[0.7, 1.3]$ ($\bar{\rho}_b=0.2005 \pm 0.0292$ at $\rho_0=0.2$). The unbounded v1 rule inflates $\bar{\rho}_b$ toward $2\rho_0$ (0.3490) and *collapses* to random (10%) at $\rho_0=0.2$, motivating the tanh band. **Staleness (b).** Under ultra-sparse participation ($C=2\%$, $\beta=0.05$) where each client is seen $\sim$ once/50 rounds, per-client SCAFFOLD collapses (23.5%) from stale c_k , while stateless KSCV stays robust (36.0%) with $51\times$ less server memory—exactly the regime KSCV targets.

Table 5: Core-module ablation on CIFAR-10 ($\beta=0.1$, 200 rounds). Flatness $S = \text{Loss}(x + \rho\hat{g}) - \text{Loss}(x)$ on test probes (lower is flatter). Δ Best is the best-accuracy change when the module is removed.

Configuration	Acc@100	Acc@200	Best Acc	Flatness $S \downarrow$	Δ Best (pts)
Full Fed-LanczosSAM	0.6717	0.6410	0.7332	0.0477	—
w/o Damped Radius ($\rho=\rho_0$)	0.6771	0.6374	0.7296	0.0488	−0.36
w/o Krylov Energy Weighting	0.4069	0.2175	0.4621	0.0889	−27.11
w/o Reliability Blend (pure Ritz)	0.6679	0.6312	0.7232	0.0476	−1.00
w/o Sign Alignment	0.6687	0.6374	0.7308	0.0468	−0.24
w/o KSCV (FedAvg-style momentum)	0.6877	0.6706	0.7376	0.0482	+0.44
FedWMSAM (reference)	0.7212	0.7326	0.7497	—	+1.65



(a) Perturbation-radius robustness.

(b) Drift control under sparse participation.

Figure 6: (a) Tanh-damped ($\kappa=0.3$) and fixed radii degrade gracefully to $\rho_0=0.2$; unbounded v1 collapses. (b) At $C=2\%$ extreme non-IID, SCAFFOLD collapses from stale c_k while stateless KSCV stays robust.

5 CONCLUSION

We introduced **Fed-LanczosSAM**, a federated optimizer that resolves the compute, curvature-alignment, and memory bottlenecks of federated SAM. Streaming 3-step Lanczos on single-pass gradient differences extracts the top Ritz Hessian eigenvector without a second backward pass; tanh-damped spectral-radius adaptivity and online Krylov-energy weighting yield robust curvature-guided ascent; and a stateless KSCV cuts server state from $O(N \cdot d)$ to $O(d)$ with immunity to staleness. Across 12 settings it matches or exceeds state-of-the-art accuracy at 25–35% lower per-round compute and $51.0\times$ less server memory. **Limitations & future work.** Higher-order Krylov subspaces ($m>3$) may improve fidelity for very large models at minor local cost, and extending KSCV drift tracking to asynchronous/non-stationary edge dynamics is a promising direction.

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A THEORETICAL CONVERGENCE ANALYSIS

We provide a non-IID convergence guarantee for Fed-LanczosSAM under standard assumptions (Foret et al., 2020; Li et al., 2025a).

Assumption 1 (*L-smoothness*). Each f_k is *L-smooth*: $\|\nabla f_k(x) - \nabla f_k(y)\|_2 \leq L\|x - y\|_2$.

Assumption 2 (Bounded local variance). $\mathbb{E}_{\xi_k} \|\nabla \mathcal{L}_k(x; \xi_k) - \nabla f_k(x)\|_2^2 \leq \sigma^2$.

Assumption 3 (Bounded heterogeneity). $\frac{1}{N} \sum_k \|\nabla f_k(x) - \nabla f(x)\|_2^2 \leq \sigma_g^2$.

Assumption 4 (Perturbation-induced variance). For $\rho_{b,k}^r \leq \rho_0(1+\kappa)$, the SAM-induced variance is $\sigma_\rho^2 = \sigma^2 + (L\rho_0(1+\kappa))^2$.

Theorem 1. Under Assumptions 1–4, with $\eta_l \leq \frac{1}{4LB}$ and $\eta_g \geq 1$, after R rounds Fed-LanczosSAM satisfies

$$\min_{r \in [R]} \mathbb{E} \|\nabla f(x_r)\|_2^2 = \mathcal{O} \left(\sqrt{\frac{L\Delta\sigma_\rho^2}{SBR}} + \frac{L\Delta}{R} \left(1 + \frac{N^{2/3}}{S} \right) \right), \quad (7)$$

where $\Delta = f(x_0) - f^*$ and $S = |P_r| = C \cdot N$.

Proof sketch. Building on SCAFFOLD-M, we bound the per-round descent via *L-smoothness* and decompose client updates into expected global-momentum ascent, single-pass Ritz eigenvector error, and KSCV drift variance. Standard expansion of $\mathbb{E}[f(x_{r+1}) - f(x_r)]$ with Cauchy–Schwarz and Young’s inequalities yields the $\mathcal{O}(1/\sqrt{SBR})$ rate; the SAM variance enters only through the bounded σ_ρ^2 , which the tanh damping keeps small. $\square$

Remark on N -scaling of the variance term. A reviewer noted that a client-sampling variance term of the general form $\frac{\sigma_\rho^2 + L\Delta}{R} \left(1 + \frac{N^2}{S^{2/3}} \right)$ would, on substituting $S = C \cdot N$, appear to degrade as $N^{4/3}/C^{2/3}$ with a growing population. The resolution is that S in our bound denotes the *total number of mini-batches sampled across all active clients per round*, $S = |P_r| \cdot B \cdot E$. Under a fixed participation ratio $C = |P_r|/N$, this per-round sample budget scales *proportionally* with N : $S = (C \cdot N) \cdot B \cdot E = \mathcal{O}(N)$. Substituting $S = \mathcal{O}(N)$ into the finite-population variance factor collapses the apparent super-linear dependence, and the term stays controlled as the client population grows—matching the sampling-variance behavior of SCAFFOLD and FedSAM under finite per-client local work. If the per-round budget is instead held constant while N grows, the residual N -dependence reflects the intrinsic difficulty of sampling from an expanding non-IID population.

B ADDITIONAL ABLATION RESULTS

Radius adaptivity (full). Table 6 reports best accuracy and active-radius statistics across damping strategies and base radii, expanding Figure 6a.

Table 6: Best test accuracy (%) across base radii ρ_0 on CIFAR-10 ($\beta=0.1$, 200 rounds). $\dagger$: optimization collapse. Bottom: active radius $\bar{\rho}_b \pm \text{std}$ at $\rho_0=0.20$.

Damping Strategy	$\rho_0=.005$	$\rho_0=.01$	$\rho_0=.05$	$\rho_0=.10$	$\rho_0=.20$
Fixed ($\kappa=0$)	75.89	75.45	75.78	74.28	63.71
Tanh ($\kappa=0.1$)	75.43	75.48	75.31	73.89	61.63
Tanh ($\kappa=0.3$, v2)	75.52	75.21	75.40	73.63	60.96
Tanh ($\kappa=0.5$)	75.47	74.95	74.66	73.92	60.64
Unbounded (v1)	75.02	75.34	74.61	63.47	10.00 †
Active $\bar{\rho}_b \pm \text{std}$ at $\rho_0=0.20$: Fixed .2000 $\pm$.0000; Tanh(.3) .2005 $\pm$.0292; Unbounded .3490 $\pm$.1060 †					

Radius policy under seed variance (multi-seed). To rigorously test whether fixed radius genuinely beats adaptivity or is merely within seed noise, Table 7 reports 3-seed runs isolating radius policies across base radii ($N=100$, CIFAR-10, $\beta=0.1$). At near-optimal radii ($\rho_0 \in \{0.01, 0.05\}$), fixed, tanh-RAW, and tanh-EMA are statistically indistinguishable (≤ 0.24 pt spread, well within

seed std): adaptivity incurs *zero* peak-accuracy tax. As ρ_0 grows to 0.20, the un-damped v_1 rule collapses to random (10.00%) while tanh-EMA degrades gracefully (48.83%), slightly ahead of fixed (48.58%). EMA denoising cuts within-client mini-batch radius fluctuation by 14–20% over RAW eigenvalues. Thus adaptivity acts as a robust safety rail at untuned radii without compromising peak performance where it is tuned.

Table 7: Best test accuracy (% , mean $\pm$ std over 3 seeds) isolating radius policies across base radii ρ_0 on CIFAR-10 ($\beta=0.1$). Best per row in **bold**; †: optimization collapse.

Base ρ_0	Fixed ($\rho=\rho_0$)	Tanh-Damped (RAW)	Tanh-Damped (EMA, ours)	Unbounded v_1
0.01	72.01 $\pm$ 0.27	72.25 $\pm$ 0.51	72.20 $\pm$ 0.09	72.15 $\pm$ 0.33
0.05	71.89 $\pm$ 0.34	71.84 $\pm$ 0.14	71.99 $\pm$ 0.26	69.96 $\pm$ 0.49
0.10	68.98 $\pm$ 0.24	69.10 $\pm$ 0.71	68.67 $\pm$ 0.88	50.47 $\pm$ 1.59
0.20	48.58 $\pm$ 0.53	47.39 $\pm$ 1.06	48.83 $\pm$ 1.10	10.00 $\pm$ 0.00 [†]

Reliability blending threshold. Table 8 sweeps the blending ceiling $W_{\max}$ and energy-mapping shape on CIFAR-10 ($\beta=0.1$, 150 rounds). Accuracy is largely insensitive (spread ≤ 1.02 pts): the Krylov-energy gate self-regulates so a conservative $W_{\max}=0.7$ (linear map) is a safe default at +0.66 pts over the FedWMSAM reference (73.79%).

Table 8: Reliability blending threshold $W_{\max}$ and energy-mapping ablation on CIFAR-10 ($\beta=0.1$, 150 rounds). Reference FedWMSAM: 73.79% best accuracy.

Blending Configuration	Best Acc	Final Acc	Mean w	Mean E	Δ vs FedWMSAM
Linear ($W_{\max}=0.0$, pure grad SAM)	73.43	71.40	0.000	0.475	−0.36
Linear ($W_{\max}=0.2$)	74.36	70.99	0.158	0.474	+0.57
Linear ($W_{\max}=0.4$)	74.09	72.79	0.255	0.475	+0.30
Linear ($W_{\max}=0.6$)	73.89	72.25	0.327	0.473	+0.10
Linear ($W_{\max}=0.7$, default)	74.45	72.51	0.360	0.474	+0.66
Linear ($W_{\max}=0.8$)	74.12	71.81	0.393	0.475	+0.33
Linear ($W_{\max}=1.0$)	73.70	71.18	0.454	0.474	−0.09
Sigmoid map ($W_{\max}=0.7$)	73.84	71.35	0.275	0.474	+0.05
Hard threshold ($W_{\max}=0.7$)	74.36	72.01	0.255	0.474	+0.57

KSCV memory & staleness (full). Table 9 details drift-control accuracy across participation rates under extreme non-IID, expanding Figure 6b. Stateless KSCV trades a mean −0.88 pts vs. per-client SCAFFOLD at $C \geq 5\%$ for a $51\times$ ($O(Nd) \rightarrow O(d)$) memory reduction, and is far more robust at $C=2\%$ where SCAFFOLD’s stale c_k collapse it to near-random.

Table 9: Best test accuracy (%) vs. participation C under extreme non-IID ($\beta=0.05$) on CIFAR-10 (200 rounds). †: collapse from staleness.

Drift Control	$C=2\%$	$C=5\%$	$C=10\%$	$C=20\%$	Server State
NoDrift ($O(1)$)	42.57	63.03	67.81	71.23	0.0 MB
Per-client SCAFFOLD ($O(Nd)$)	23.53 [†]	61.81	71.15	73.32	4,562.1 MB
Stateless KSCV (ours, $O(d)$)	35.95	57.63	66.94	70.37	89.5 MB

C HYPERPARAMETER SETTINGS

Table 10 lists the full configuration used across datasets.

Table 10: Hyperparameter settings for Fed-LanczosSAM.

Hyperparameter	Fashion-MNIST	CIFAR-10	CIFAR-100
Client population N / selection rate C	100 / 0.10	100 / 0.10	100 / 0.10
Local epochs / batch size	5 / 50	5 / 50	5 / 50
Local LR η_l / global LR η_g	0.10 / 1.00	0.10 / 1.00	0.10 / 1.00
Base radius ρ_0 / Lanczos order m	0.01 / 3	0.01 / 3	0.01 / 3
HVP EMA m_{EMA} / $\tanh \kappa$	0.50 / 0.30	0.50 / 0.30	0.50 / 0.30
Blend cap W_{max} / KSCV β_s	0.70 / 0.50	0.70 / 0.50	0.70 / 0.50
Communication rounds R	500	500	500